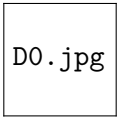
Lecture 30: Functional Dependencies

Z2004: Database Management Systems

Dr. Innocent Nyalala



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Outline

- The messy table problem
- Three anomalies
- What is a Functional Dependency?
- FD examples from our table
- Trivial vs. non-trivial FDs
- Attribute closure
- Finding candidate keys
- Think-Pair-Share
- Summary

Module 7 — Normalisation

This lecture opens the normalisation unit. FDs are the **foundation** — everything in L31–L33 builds on today.

The Messy Table — A Disaster Waiting to Happen

SID	SName	SEmail	CID	CName	IID	IName	Grade	Sem
S001	Amani	amani@st	CS101	Databases	I01	Dr. Ouma	A	Sp26
S002	Baraka	baraka@st	CS101	Databases	I01	Dr. Ouma	B	Sp26
S003	Chidi	chidi@st	CS202	Algorithms	I02	Dr. Said	A	Sp26
S001	Amani	amani@st	CS202	Algorithms	I02	Dr. Said	C	Sp26
S004	Dina	dina@st	CS101	Database	I01	Dr. Ouma	B+	Sp26
S005	Emil	emil@st	CS202	Algorithms	I03	Dr. Said	A-	Sp26

"Databases" vs "Database"!

Duplicated 3×!

This table is a disaster waiting to happen

9 attributes, endless redundancy. InstructorName stored in every row. One typo creates inconsistency.

Three Anomalies of Bad Design

Insertion Anomaly

Want to add new course **CS303: OS**?

⚠ *No student enrolled yet — can't insert!*

CourseID and CourseName trapped inside a student row.

Deletion Anomaly

Chidi (S003) drops CS202 (only student)?

⚠ *We lose CS202 and Dr. Said's info too!*

Course data tied to student enrolment rows.

Update Anomaly

Dr. Ouma changes name to Dr. Abubakar?

⚠ *Must update 3 rows. Miss one = corrupt!*

Every course section stores instructor name.

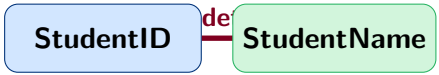
Root Cause

Data that **belongs together logically** is scattered across multiple rows. The fix: understand **Functional Dependencies** and use them to split tables properly.

What Is a Functional Dependency?

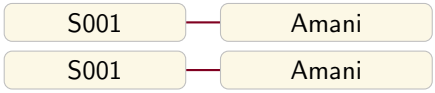
Definition

$X \to Y$ means: **knowing X uniquely determines Y**.
For every pair of tuples t_1, t_2:
if $t_1[X] = t_2[X]$ then $t_1[Y] = t_2[Y]$



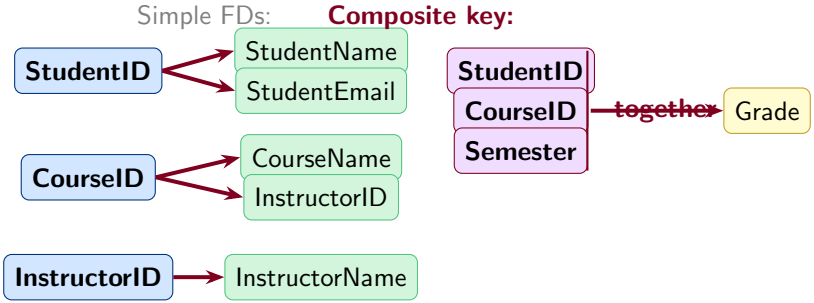
Intuition: X is the "key", Y is the "value"

- **StudentID** $\to$ **StudentName**
  Same ID $\Rightarrow$ same name. Always.
- **CourseID** $\to$ **CourseName**
  Same course code $\Rightarrow$ same name.
- **Email** $\not\to$ **StudentID**?
  Two students could share an email (family!)



✔ **Same ID $\Rightarrow$ Same name**

Functional Dependencies in `StudentEnrolment`



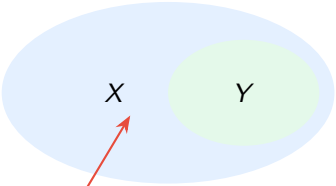
Key Insight

Grade needs all three — knowing only StudentID or only CourseID is not enough.

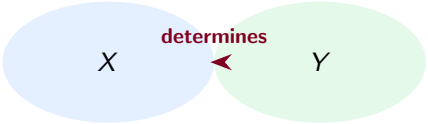
Why This Matters

These FDs tell us **how to split the table**. Each FD group → its own relation.

Trivial vs. Non-Trivial Functional Dependencies



TRIVIAL FD
$X \to Y$ where $Y \subseteq X$
Always true, says nothing.
e.g. {SID, CID} → SID
Of course! SID is in the set.



NON-TRIVIAL FD
$X \to Y$ where $Y \not\subseteq X$
Says something meaningful!
e.g. CourseID → CourseName
One key, different attribute.

Rule of Thumb

In normalisation we only care about **non-trivial FDs**. Trivial ones are always satisfied and tell us nothing about the schema.

Attribute Closure: What Can We Infer?

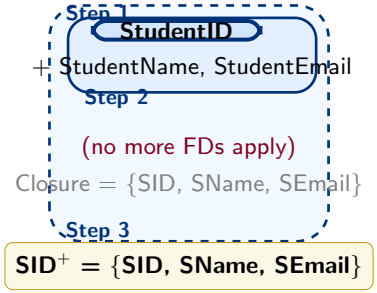
Question: Starting with {StudentID}, what can we determine?

Given FDs:

- StudentID → StudentName, StudentEmail
- CourseID → CourseName, InstructorID
- InstructorID → InstructorName
- {SID, CID, Sem} → Grade

Algorithm (*closure*$^+$ of X):

1. Start: result = X
2. Scan all FDs: if left side $\subseteq$ result, add right side
3. Repeat until no change



Notation

{StudentID}$^+$ = all attributes determined by StudentID (under the given FD set)

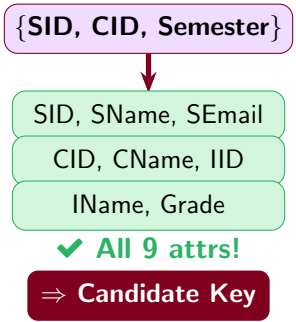
Finding Candidate Keys Using Closure

Candidate Key Test

X is a **superkey** if $X^+ =$ all attributes.
X is a **candidate key** if it is a **minimal** superkey (remove any attribute and it no longer determines everything).

Test: {StudentID, CourseID, Semester}

1. Compute closure...
2. SID $\rightarrow$ SName, SEmail
3. CID $\rightarrow$ CName, IID
4. IID $\rightarrow$ IName
5. {SID,CID,Sem} $\rightarrow$ Grade
6. **Result:** all 9 attributes ✔



Think-Pair-Share (4 minutes)

Given these FDs — find all candidate keys

Relation `R(A, B, C, D, E)`

- $A \rightarrow BC$
- $CD \rightarrow E$
- $B \rightarrow D$
- $E \rightarrow A$

- Compute closure of {A}:

  _____

- Does {A} determine everything?
- Try other sets.
- Which sets are **minimal**?

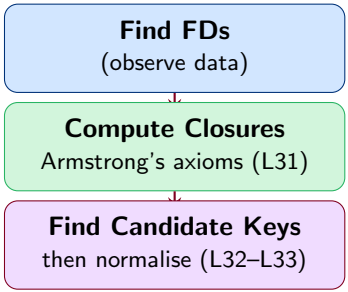
Hint

There may be **more than one** candidate key!

Summary

Key Concepts Today

- **Bad design** $\Rightarrow$ 3 anomalies
- $X \rightarrow Y$: knowing X gives Y
- **Trivial** FD: $Y \subseteq X$ (boring)
- **Non-trivial**: meaningful dependency
- **Closure** X^+: all attrs from X
- **Candidate key**: minimal superkey



Next Lecture — L31: Armstrong's Axioms

Formal inference rules to derive *all* valid FDs from a given set. Foundation of the canonical cover algorithm.